UI/UX CONTENT & FLOW SPECIFICATION

# Grid-Guard AI

## Screens, Elements & Navigation — Design-Tool Input

*Functional content only: screens, on-screen elements, states, and navigation flow. No visual design, layout, or styling decisions — those are made separately in the design tool.*

| Field | Value |
|---|---|
| Author | Noor Eman |
| Version | 1.0 (Draft) |
| Purpose | Input document for Figma or any design tool — lists what each screen needs to contain and how screens connect, not how they should look |

# 1. How to Use This Document

Each screen below is specified as: its purpose, the data/content it must display, the actions a user can take on it, where it's navigated from and to, and any non-default states it must handle (empty, loading, error). No colors, spacing, typography, or layout composition are specified — those decisions belong entirely to the design process.

# 2. Overall Navigation Map

## 2.1 Web Dashboard

```
Login
  └→ Risk Queue (landing screen)
        ├→ Consumer Detail ⟶ Assign Case (modal/panel)
        ├→ Case List ⟶ Case Detail (read-only, links to Audit Log)
        ├→ Feeder Map
        └→ Model Performance Panel
  Audit Log is reachable from Case Detail, not top-level nav.
```

## 2.2 Mobile App

```
Login
  └→ My Cases (landing screen)
        └→ Case Detail
              └→ Inspection Form ⟶ (on submit) back to My Cases
  Sync Status / History reachable from My Cases.
```

# 3. Web Dashboard — Screens

## 3.1 Login

- Purpose: authenticate an officer or circle manager.
- Content: email field, password field, submit action, error message area.
- States: default; submitting; error (invalid credentials — show inline message, do not specify which field is wrong).
- Navigation: on success → Risk Queue.

## 3.2 Risk Queue (landing screen)

- Purpose: primary daily view — ranked, filterable list of at-risk consumers.
- Content per list row: consumer reference/ID, address, feeder name, risk score (numeric), risk band (high/medium/low as a distinct, scannable indicator), meter type.
- Filters available: feeder (dropdown/select), risk band (select), meter type (select). Filters can combine.
- Actions: click/tap a row → go to Consumer Detail. Pagination control if list exceeds one page.
- States: loading (initial fetch); empty (“no consumers match current filters”, distinct from “no data at all”); error (fetch failed, with a retry action).
- Persistent element: a visible “demo/benchmark data” indicator, since this version does not use live DISCO data (per PRD data-honesty requirement) — must be visible on this screen, not just buried in settings.

- Navigation: entry point after login; each row → Consumer Detail.

### 3.3 Consumer Detail

- Purpose: let an officer inspect one flagged consumer closely before deciding to assign a case.
- Content: consumer ID, address, feeder, meter type; risk score and band; plain-language explanation text (e.g. “consumption dropped 60%...”); a consumption trend visualization (time series of usage over recent periods).
- Actions: “Assign Case” action, opening the Assign Case panel/modal; back/close to return to Risk Queue.
- States: loading; error (consumer not found).
- Navigation: entered from Risk Queue; “Assign Case” opens 3.4.

### 3.4 Assign Case (modal or side panel, not a full screen)

- Purpose: assign the currently-viewed consumer as a case to a specific inspector.
- Content: inspector selector (list of available inspectors by name), optional free-text note field, confirm/cancel actions.
- States: submitting; success (confirmation, then close and return to Consumer Detail or Risk Queue); error (e.g. inspector has no active session / assignment failed).
- Navigation: opened from Consumer Detail; closes back to the screen it was opened from.

### 3.5 Case List

- Purpose: track all cases and their current status.
- Content per row: consumer reference, assigned inspector name, status (assigned / in progress / closed — confirmed / closed — false positive / closed — inconclusive), date created.
- Filters: status, feeder.
- Actions: click/tap a row → Case Detail (read-only view).
- States: loading; empty (no cases yet, or none matching filters); error.
- Navigation: reachable from main nav; each row → Case Detail.

### 3.6 Case Detail (read-only)

- Purpose: full record of a single case, including its outcome once closed.
- Content: consumer info (address, feeder, meter type), original risk score/explanation, assignment info (who, when, note), and — once closed — the inspection outcome, tampering-type checklist selections, note, photo, and GPS location.
- Actions: link/button to view full Audit Log for this case.
- States: pending (not yet inspected — outcome section shows “awaiting inspection” rather than blank); closed (full outcome shown).
- Navigation: entered from Case List; links to Audit Log (3.8).

### 3.7 Feeder Map

- Purpose: geographic view of where flagged/case consumers are concentrated.

- Content: map with markers per flagged consumer, colored/marked by risk band; clicking a marker shows a brief summary (consumer ref, risk band) with a link to Consumer Detail.
- Filters: same as Risk Queue (feeder, risk band) so the map and list stay conceptually consistent.
- States: loading; empty (no geocoded consumers to show).
- Navigation: reachable from main nav; marker click → Consumer Detail.

### 3.8 Model Performance Panel

- Purpose: let an officer or circle manager judge whether flags are actually useful, using closed-case outcomes.
- Content: confirmed-theft rate, false-positive rate, and inconclusive rate among closed cases (as percentages); a trend view of confirmed-rate over recent months.
- States: loading; empty (no closed cases yet to compute rates from — must not show a misleading 0% or blank chart without this label).
- Navigation: reachable from main nav.

### 3.9 Audit Log

- Purpose: full, immutable history of state changes for a case (or system-wide, if accessed outside a case context).
- Content per entry: action description (plain language, e.g. “Case assigned to Bilal Ahmed”), actor name (or “system” for automated flag generation), timestamp — shown as a chronological list, oldest or newest first (decide in design phase, not specified here).
- States: loading; empty (should not normally occur once a case exists, since flag_generated is always the first entry).
- Navigation: reached from Case Detail.

## 4. Mobile App — Screens

### 4.1 Login

- Purpose: authenticate an inspector.
- Content: email field, password field, submit action, error message area.
- States: default; submitting; error.
- Navigation: on success → My Cases. Session should persist across app restarts (no re-login required daily).

### 4.2 My Cases (landing screen)

- Purpose: the inspector's personal queue of assigned cases.
- Content per row: consumer address (primary identifier for a field worker, more useful than an abstract ID), risk band, case status, sync status indicator (synced / pending).
- Sort: priority/risk band first, then date assigned — exact ordering logic decided in design/dev, but risk should visibly drive order.
- Actions: tap a case → Case Detail.

- States: loading; empty (“no cases assigned”); offline (queue still shown from local cache — must work fully without connectivity, per MOB-5).
- Navigation: entry point after login; each row → Case Detail; also reachable: Sync Status/History.

### 4.3 Case Detail

- Purpose: give the inspector what they need before and during a site visit.
- Content: consumer address, meter ID, plain-language risk reason (same explanation text as the web dashboard's Consumer Detail), any assignment note from the officer.
- Actions: “Start Inspection” → opens Inspection Form.
- States: works fully offline once cached locally.
- Navigation: entered from My Cases; “Start Inspection” → 4.4.

### 4.4 Inspection Form

- Purpose: record the outcome of an on-site inspection.
- Content/fields: outcome selector (confirmed theft / honest / inconclusive — mutually exclusive choice); tampering-type checklist (multi-select: e.g. bypass wire, meter reversed, remote-control device — exact list finalized with domain input, not fixed here); free-text note field; photo-capture action (opens device camera); GPS coordinates (captured automatically on form open or on submit, not manually entered).
- Actions: submit (writes locally immediately, regardless of connectivity); cancel/back (discards unsaved entry, with a confirmation prompt if fields are filled).
- States: unsaved/in-progress; submitted-pending-sync; submitted-synced. The form itself does not need a distinct “offline” state — it behaves identically online or offline (this is the point of MOB-5).
- Navigation: entered from Case Detail; on submit → back to My Cases, with the case now reflecting updated status.

### 4.5 Sync Status / History

- Purpose: let the inspector see what has and hasn't synced, and review past closed cases.
- Content: list of past submissions with sync status (pending / synced) and, for synced ones, timestamp of sync; separate section or filter for fully closed case history.
- Actions: manual “retry sync” action for pending items, in addition to automatic background sync.
- States: empty (no submissions yet); all-synced (no pending items — should read as a clearly positive/complete state, not just an empty list).
- Navigation: reachable from My Cases.

## 5. Shared Content Notes (Both Surfaces)

- Risk band terminology (“high / medium / low”) and any color-independent indicator (e.g. also using text/icon, not color alone) must be consistent between web and mobile — an officer and inspector discussing the “same” case should see the same language.

- Explanation text (from the ML output schema) is written to already be plain-language at the API level — screens should display it as provided, not attempt to further summarize or truncate it in ways that change its meaning.
- Mobile app text should be available in both English and Urdu (per PRD MOB-6) — all screens in Section 4 need their content authored with translation in mind (avoid embedding text in images, avoid concatenated strings that don't translate cleanly).
- No screen anywhere should present the demo/benchmark data as live production data — the Risk Queue's data-source indicator (3.2) is the primary place for this, but any screen showing consumer data inherits the same constraint.